Lesson notes · for you

Hello, Python

Programming · Level 2 · Lesson 1

Time
20 min

Materials
A phone with this lesson downloaded; A notebook for predictions

Learning goals

- Print text and numbers.
- Calculate with + - * / and **.
- Run a program top to bottom, exactly as written.

Activity: Predict, then run

1. Warm-up (3 min): Without running it, write what print(2 + 3 * 4) shows. Compare with a partner.
2. Predict (7 min): Write down what each prints: print("7 + 3"), print(7 + 3), print((2 + 3) * 4), print(2 ** 3) and print("Total:", 5 * 4).
3. Run (7 min): Run each one on the phone, or in the trace lab, and tick or cross your predictions.
4. Explain (3 min): Pick one prediction you got wrong and explain why Python did what it did.

You can do the predictions on paper with no phone, then check them together on one phone.

Discussion

- Why does Python show text exactly as written?
- When do you need brackets in a sum?
- What is an error message good for?

Check yourself

- Write a line that prints 10, and a line that prints 7 + 3 as text.

Try it this week
Write a three-line program that prints your name, how old you will be in 10 years, and 2 ** 10. Predict each line before you run it.

Worksheet

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Name _____ Class _____ Date _____

1 **What does this line show?**

```
print("10 * 2")
```

◯ 10 * 2, as written ◯ An error ◯ 20

2 **The first two lines ask for a classroom's length and width in metres. Print its area and its perimeter, like Area: 48 and Perimeter: 28, each on its own line.**

```
length = int(input("Length? "))
width = int(input("Width? "))
# print the area, then the perimeter
```

Write your program:

3 **Put the lines in order so the program prints exactly: Market receipt, Mangoes: 150, Bread: 240, Total: 390. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `print("Total:", 3 * 50 + 2 * 120)`

☐ `print("Bread:", 2 * 120)`

☐ `print("Market receipt")`

☐ `print("Total:", "3 * 50 + 2 * 120")`

☐ `print("Mangoes:", 3 * 50)`

4 **print("7 + 3") shows 10.**

◯ True ◯ False

5 **Match each sum to what Python gives.**

2 + 3 * 4	• • • •	8
(2 + 3) * 4	• • • •	3.5
2 ** 3	• • • •	14
7 / 2	• • • •	20

What would you like your first useful program to do? Write what it should print, line by line.

Check your answers

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1 **10 * 2, as written**

In quotes it is text, so Python shows it exactly as written.

2

```
length = int(input("Length? "))
width = int(input("Width? "))
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

Area is length × width. The perimeter needs brackets, 2 * (length + width), because Python multiplies before it adds.

3 **Top to bottom: 4, 3, 1, X, 2**

```
print("Market receipt")
print("Mangoes:", 3 * 50)
print("Bread:", 2 * 120)
print("Total:", 3 * 50 + 2 * 120)
```

A program runs top to bottom, so the lines go in the order you want to see them.

4 **False**

The quotes make it text, so Python shows 7 + 3 exactly as written.

5

2 + 3 * 4 — 14

(2 + 3) * 4 — 20

2 ** 3 — 8

7 / 2 — 3.5

* comes before +, brackets come first of all, ** is a power (2 × 2 × 2), and / gives a decimal.